

The Gradient of the Logistic Function

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September 12, 2026

1 What does a Logistic Function do?

Take one row of observations. Let there be d observations and a classification y where $y \in \{0, 1\}$. You can think of y as yes or no, or true or false. Let each of the d observations we represented by a variable x_j . Where each x_j is the z-score for observation (the standard deviations from the mean).

Our goal is simply take these x_j and predict y as best as possible. So intuitively we can take a single observation x_2 say and whenever its > 1 say we let our prediction by 1 for y . But this doesn't take into account all our observations. So let sum them, let $z = x_1 + x_2 + \dots + x_d + b$. We introduced a b which is a bias for the guess we want. You can imagine b as if all x_j are 0 then what should our guess be.

Now you might think now that some observations may be more important than others or interact with y in different ways. For example your length of credit history might be more important than your hair color in predicting if you will repay or not.

So assign a weight w_j to each observation x_j . So we have now $z = w_1x_1 + w_2x_2 + \dots + w_dx_d + b$. (Make these equations).

But how do we get these weights? We need to say based on these current weights how confident are we that $y = 1$ ie given this value of z how confident are we that this observation as $y = 1$. Well z is arbitrary, w_1 can be as big as we want z can be 100, -0.5 so on; what does that mean?

Take a function $\frac{1}{1+e^{-z}}$ and let this be how predicted probability that $y = 1$. Specifically let $p = \sigma(z) = \frac{1}{1+e^{-z}}$. Then observe that for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that whenever $z \geq N$ $|\frac{1}{1+e^{-z}} - 1| < \epsilon$, specifically choose

$z > \log(\frac{1}{\epsilon})$. To see this $|\frac{1}{1+e^{-z}-1}| = |\frac{1}{1+e^{-z}} - \frac{1+e^{-z}}{1+e^{-z}}| = |\frac{-e^{-z}}{1+e^{-z}}| = \frac{e^{-z}}{1+e^{-z}}$ since $e^x > 0$ always we have $1+e^{-z} > 1$ giving $\frac{e^{-z}}{1+e^{-z}} < e^{-z}$ now $e^{-z} < \epsilon = -z < \log(\epsilon) = z > -\log(\epsilon) = z > \log(\frac{1}{\epsilon})$. Meaning whenever $z \geq N$ where $N > \log(\frac{1}{\epsilon})$ we have $|\frac{1}{1+e^{-z}-1}| < \epsilon$ giving $p - 1 \leq \epsilon$.

For the other direction $|\frac{1}{1+e^{-z}-0}| < \epsilon$, (Finish later)

Now that we have proved $p \in (0, 1)$ and we have defined it as the probability that $y = 1$ for our model we can define the loss that we are looking to minimize. Intuitively we want our probability of $y = 1$ distance from the true value. But an absolute difference doesn't account for the confidence difference we between a precision of .99 and .999 which are 0.01 and 0.001 away from 1 but are dramatically different in confidence. Therefore we define the loss using logs. Specifically we let the loss = $-\log(p)$ if $y = 1$ and $-\log(1 - p)$ if $y = 0$. In closed form our loss looks like $L = -[y \log(p) + (1 - y) \log(1 - p)]$.

As stated before we want to minimize our loss but we want to tune our arbitrary weights. So how can we do this. We want to tune our weights as optimally as possible. Therefore we use the gradient specifically $\frac{\partial L}{\partial w_j}$. But we defined L in terms of y and p so we must chain through to get our derivative.

(To continue derive the chain rule, then explain b and using 1 at the end of the matrix for the $x_d + 1$, ie because the x_j comes out and b's will be 1 always)

(After this reformat and its done)